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Plan for the ground you will drive on, not the ground you see today

Forecast-aware off-road route planning for military vehicles, computed on the vehicle with no weather service and no network link

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Poster track
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Routes planned on a forecast made **on the vehicle** sent **24% fewer** trips into mud than routes planned on **today's map**, three days out, and held back fewer that could have gone. At the same risk, they hold back **31–40% fewer** trips than a safety margin on today's map.

12 ms per 7-day forecast, one CPU core
18 B to send it to a convoy
2.0 MB model size, 531k parameters
25 ms to re-plan a route on a 76k-cell map

1 **Weather log** — rain + 2-layer soil moisture, one 9 km cell, daily
→ 2 **Edge forecaster** — GRU 128×2 + ridge, RCI change, 1–7 days
→ 3 **Safety margin** — one per firmness band, tuned on the other year
→ 4 **Soil map** — RCI spread by terrain wetness index, 90 m
→ 5 **Fuzzy AND** — min(slope, cover, soil), Liu et al. tables [1]
→ 6 **Route planner** — fastest route, no NO-GO, convoy shares its tracks

all on the vehicle: no weather service, no network link, one CPU core

01 The ground changes faster than the map

Off the road, the soil decides who moves. On the Pachmarhi plateau a monsoon week takes ground firmness from over 200 psi to under 20, so a route that is GO when planned can be **NO-GO** when the column arrives.

Maps describe the ground **today** [1]. We plan for the day of travel.

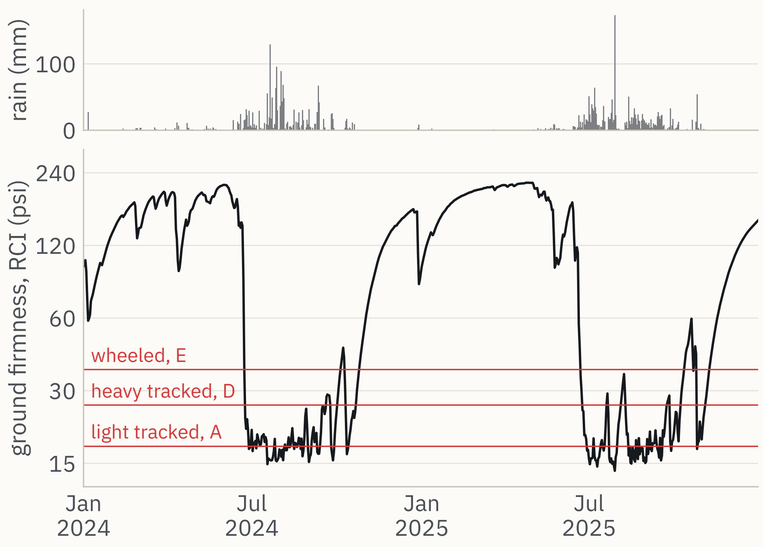


Fig. 1 Rain and ground firmness (rating cone index, RCI), 2024–25. Below its red line, a vehicle band is stuck.

02 Firmness from soil moisture, limits from the vehicle

$SWI = 0.35\ s_{0–7\ cm} + 0.65\ s_{7–28\ cm}$

$RCI = 300 \cdot e^{-3.0\ SWI} \cdot (1 - 0.15 \min(R_{3d} / 50\ mm, 1))$

NO-GO if $RCI < VCI_1$ · SLOW-GO if $< VCI_{50}$ · GO otherwise

s: ERA5-Land soil moisture, scaled 0 (driest) to 1 (wettest) in 11 years. The wet end, 15 psi, follows field readings where a 4×4 bogged down [2].

Each of 1,145 ground vehicles gets its VCI_1 and VCI_{50} from the NRMM Mobility Index [2, 3]. Our tracked version reproduces the published M1 example.

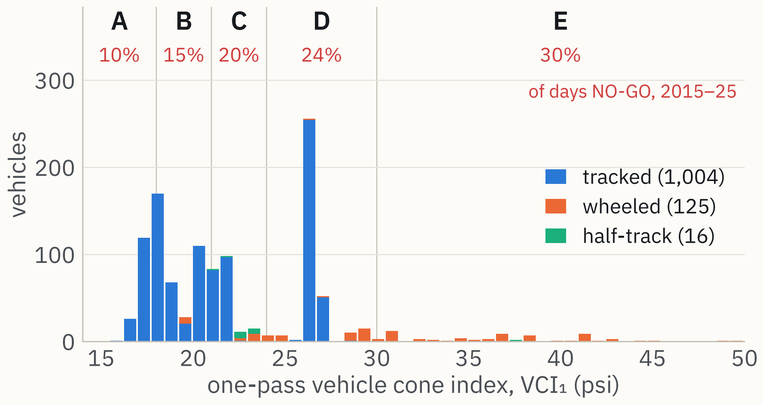


Fig. 2 The fleet in five firmness bands: light tracked (A) stuck on 10% of days, wheeled (E) on 30%.

What went in

Weather	ERA5-Land soil moisture; IMD rain (ERA5 in 2025); daily 2015–25
Terrain	Earth Engine: DEM, WorldCover, Sentinel-2; 90 m grid, 76k cells
Roads	OpenStreetMap, 403 usable segments
Vehicles	1,004 tracked, 125 wheeled, 16 half-tracks

03 A forecast that fits on the vehicle

A GRU and a ridge regression predict the **change** in RCI 1–7 days ahead from the local weather log alone: 20% lower error than "tomorrow equals today" at 7 days.

A safety margin per band, tuned on one test year and scored on the other, turns it into a NO-GO call.

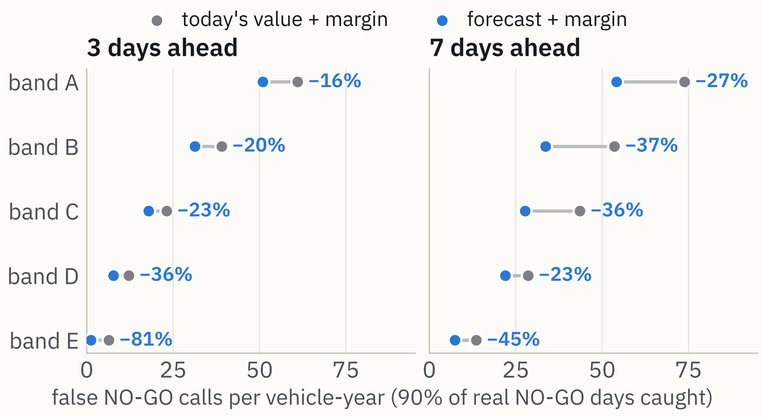


Fig. 3 Same 90% of NO-GO days caught, 23–45% fewer false alarms at 7 days: fewer days a unit stays parked for nothing.

04 What it cannot see

Without a weather forecast, nothing predicts the day the monsoon breaks. The gain comes in the days around it.

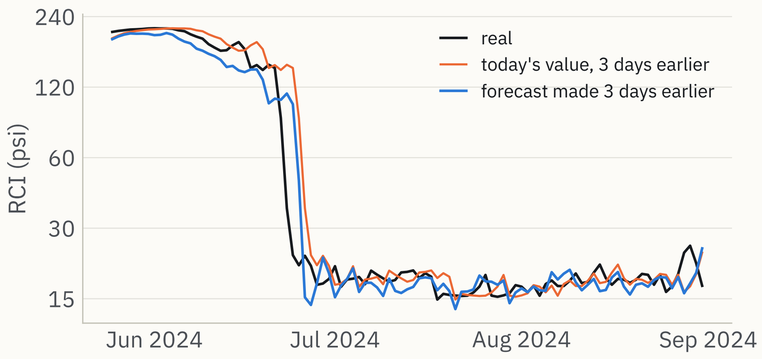


Fig. 4 Monsoon onset, 2024: firmness three days ahead.

05 Same trip, two plans

Fig. 5 A tracked vehicle from band D (VCI_1 26 psi) plans on 15 June 2025 for a trip 7 days later. On the planning day the ground reads 128 psi and the direct route (orange) looks open. On the travel day it reads 34 psi and that route crosses NO-GO soil (×). The route planned on the forecast (blue, margin 19.5 psi) avoids it. Circle: start; square: end; grey lines: OpenStreetMap roads.

06 Two monsoons, 325,800 trips

Every day of 2024–25, 30 trips of 8–25 km per band, leaving 1, 3 or 7 days after planning, checked against the real ground on the day of travel. A planner fails two ways: it sends a vehicle into mud, or holds back a trip that was possible.

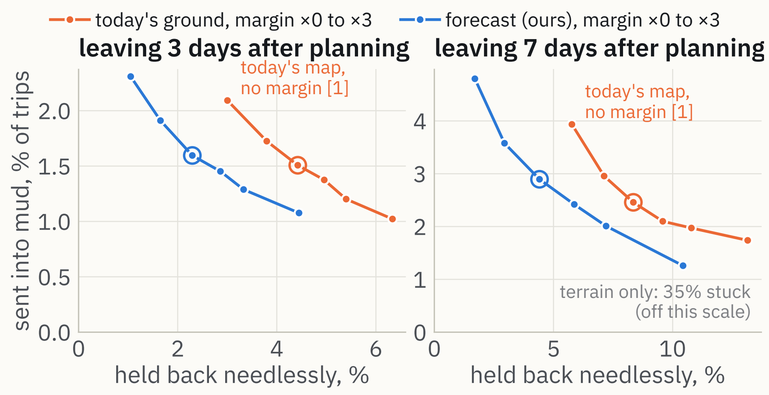


Fig. 6 Both errors as each margin goes from none to 3× the tuned value (ringed). The forecast is closer to the corner at every setting.

leaving in 3 days	stuck	held back	extra time
terrain only	35.5%	0.0%	+0.43%
today's map [1]	2.1%	3.0%	+0.06%
today + margin	1.5%	4.4%	+0.06%
forecast + margin	**1.6%**	**2.3%**	**+0.05%**

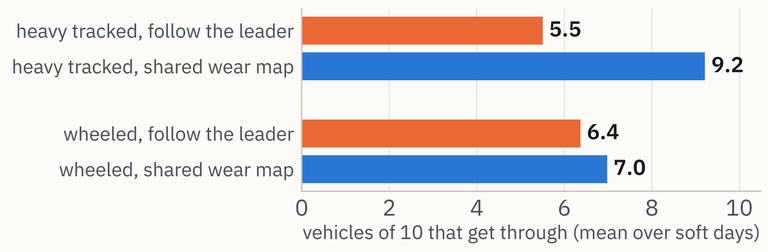
07 An average hides the deal-breaker

A weighted mean [1] can call a cell passable when one factor says no. The minimum cannot.

combining slope, cover, soil	stuck, 1 d	3 d	7 d
minimum (fuzzy AND)	**0.7%**	**1.6%**	**2.9%**
weighted mean, with veto [1]	0.9%	1.8%	3.0%
weighted mean, no veto	34.6%	34.8%	34.9%

08 Convoys: share the track, not the map

Each pass churns soft ground. Of ten heavy tracked vehicles, 5.5 get through when they **follow the leader** and 9.2 when each re-plans on a **shared pass-count map**, for 6.9 kB of track updates.



09 Send the forecast, not the map

what crosses the radio link	size
7-day firmness forecast	**18 B**
a whole convoy's tracks	6.9 kB
one passability map, 1 B per cell	76 kB
maps for 5 bands × 7 days	2.7 MB

Vehicles rebuild maps locally: the link carries 4,200× less than one map.

Limits. Ground firmness is computed from modelled soil moisture (ERA5-Land, one 9 km cell), not measured. Spreading it over the map by wetness index is a heuristic with no field check. Vehicle masses come from a game-file dataset, with wheeled ones corrected from open sources, so we report bands, not named vehicles. Tuned margins differ between the two test years by up to 7.2 psi. **Next:** cone-penetrometer readings at Pachmarhi, Sentinel-1 radar as a soil-moisture input, and ruts that last between days.

1. Liu Q. et al. Dynamic trafficability map for unmanned vehicles. Sci. Rep. 15:9957, 2025.
2. Roșca P. et al. Vehicle cone index for fine-grained soil. J. Mil. Technol. 1(2), 2018.
3. Ciobotaru T. Semi-empiric algorithm for vehicle mobility. LEJPT 8(15):19–30, 2009.
4. Muñoz-Sabater J. et al. ERA5-Land. Earth Syst. Sci. Data 13:4349–4383, 2021.
5. Zanaga D. et al. ESA WorldCover 10 m v200, 2022. Roads © OpenStreetMap contributors.

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